

Multistate Landau–Zener Solver: Augmented ODE with Gauged Dynamics

Reference Summary — v3

1 Parameters

Fix dimension $N \geq 2$. The model is specified by three vectors:

$$\gamma_i, \quad \epsilon_i, \quad a_i, \quad i = 0, \dots, N-1$$

subject to strict ordering $\epsilon_0 < \epsilon_1 < \dots < \epsilon_{N-1}$.

We fix a preferred initial eigenvector index $x \in \{0, \dots, N-1\}$.

2 Derived Quantities

All quantities are algebraic functions of u and $\lambda_0(u), \dots, \lambda_{N-1}(u)$.

2.1 Implicit eigenvalues λ_i

The $\lambda_i(u)$ satisfy:

$$\sum_{j=0}^{N-1} \frac{\gamma_j^2}{\lambda_i - \epsilon_j} = u \quad \text{for each } i = 0, \dots, N-1 \quad (1)$$

with ordering:

- $\lambda_i \in (\epsilon_{i-1}, \epsilon_i)$ for $i = 1, \dots, N-1$ (interior eigenvalues),
- $\lambda_0 < \epsilon_0$ when $u < 0$; $\lambda_0 > \epsilon_{N-1}$ when $u > 0$ (exterior eigenvalue).

The exterior eigenvalue diverges as $\lambda_0 \sim \sum \gamma_j^2 / u$ when $u \rightarrow 0$.

2.2 Sign factors δ_i

$$\delta_0 = (1 - 2\theta(u)) \operatorname{sign}(\gamma_0), \quad \delta_i = \operatorname{sign}(\gamma_i) \quad \text{for } i \geq 1 \quad (2)$$

2.3 Auxiliary sums

$$\Sigma_i^{(2)} \equiv \sum_{j=0}^{N-1} \frac{\gamma_j^2}{(\lambda_i - \epsilon_j)^2} \quad (3)$$

$$\Sigma_i^{(3)} \equiv \sum_{j=0}^{N-1} \frac{\gamma_j^2}{(\lambda_i - \epsilon_j)^3} \quad (4)$$

2.4 Energies, couplings, and coupling matrix

$$E_i = \sum_{j=0}^{N-1} \frac{a_j \gamma_j^2}{\lambda_i - \epsilon_j} \quad (5)$$

$$\Gamma_i = \frac{\delta_i}{\sqrt{\Sigma_i^{(2)}}} \quad (6)$$

$$S_{ij} = \frac{\Gamma_i \Gamma_j}{\lambda_i - \lambda_j} \quad (i \neq j), \quad S_{ii} = 0 \quad (7)$$

Note: $S_{ji} = -S_{ij}$ (antisymmetric).

2.5 Matrix View

$$H_{ij}(u) = \gamma_i \gamma_j \frac{a_i - a_j}{\epsilon_i - \epsilon_j} \quad (8)$$

$$H_{ii}(u) = a_i u - \sum_{j \neq i} \gamma_j^2 \frac{a_i - a_j}{\epsilon_i - \epsilon_j} \quad (9)$$

with eigenvalues E_j (5) corresponding to eigenvectors $(\vec{v}_j)_k \propto \frac{\gamma_k}{\lambda_j - \epsilon_k}$.

2.6 Smoothness of S across $u = 0$

Although $\lambda_0 \rightarrow \pm\infty$ and δ_0 flips sign at $u = 0$, the coupling matrix S_{ij} is smooth there. For $i = 0$, $j \geq 1$:

$$\Gamma_0 \sim \frac{\delta_0 \lambda_0}{\sqrt{\sum \gamma_k^2}} \implies S_{0j} = \frac{\Gamma_0 \Gamma_j}{\lambda_0 - \lambda_j} \rightarrow \frac{\delta_0 \Gamma_j}{\sqrt{\sum \gamma_k^2}}$$

From the $u < 0$ side: $\delta_0 = +\text{sign}(\gamma_0)$, $\lambda_0 \rightarrow -\infty$, so $\lambda_0/|\lambda_0| = -1$.

From the $u > 0$ side: $\delta_0 = -\text{sign}(\gamma_0)$, $\lambda_0 \rightarrow +\infty$, so $\lambda_0/|\lambda_0| = +1$.

Both give the same limit for S_{0j} .

This smoothness is the key structural property: it means the wavefunction equations require *no discontinuous treatment at $u = 0$* .

3 Augmented ODE System

3.1 Interior eigenvalue equations

Differentiating (1) for $i = 1, \dots, N-1$:

$$\boxed{\lambda'_i(u) = -\frac{1}{\Sigma_i^{(2)}}} \quad i = 1, \dots, N-1 \quad (10)$$

These are promoted to dynamical ODE variables.

3.2 Exterior eigenvalue λ_0

λ_0 is **not** an ODE variable: its equation $\lambda'_0 = -1/\Sigma_0^{(2)}$ has a finite-time singularity at $u = 0$ (since $\Sigma_0^{(2)} \rightarrow 0$ as $\lambda_0 \rightarrow \pm\infty$).

Instead, λ_0 is recovered at each step by solving the implicit constraint (1) via Newton iteration, using the asymptotic starting point $\lambda_0 \approx \sum \gamma_j^2/u$.

3.3 Wavefunction equations (gauged)

$$\boxed{f'_i(u) = - \sum_{j \neq i} S_{ij} f_j(u) + i(E_i - E_x) f_i(u)} \quad (11)$$

This is the Multistate Landau Zener for Type-1 matrices in the adiabatic basis. The gauge choice sets the effective frequency of f_x to zero. Since $|f_i|^2$ is gauge-invariant, all transition probabilities are unchanged.

3.4 State vector

The system has $2N-1$ complex ODEs:

$$\mathbf{y}(u) = \left(\underbrace{\lambda_1, \dots, \lambda_{N-1}}_{N-1 \text{ interior eigenvalues}}, \underbrace{f_0, \dots, f_{N-1}}_{N \text{ wavefunctions}} \right) \in \mathbb{C}^{2N-1}$$

λ_0 is computed from the constraint at each RHS evaluation; δ is computed from u on the fly.

3.5 No phase split required

Because S_{ij} is smooth across $u = 0$ (§2.6), the RHS of *all* ODE variables is continuous. The integration proceeds in a single pass from u_0 to u_f .

4 Initial Conditions at $u_0 \ll -1$

4.1 Initial $\lambda_i(u_0)$

Clear denominators in (1) to form a degree- N polynomial. Solve and assign roots to intervals per Section 2.1. The state stores only $\lambda_1, \dots, \lambda_{N-1}$.

4.2 Initial $f_i(u_0)$

For the preferred eigenvector x :

$$f_{x;j} = S_{xj} \left(\frac{i}{E_x - E_j} + \frac{\Gamma_x - \Gamma_j}{\lambda_x - \lambda_j} - \Gamma_x^2 \Sigma_x^{(3)} - \Gamma_j^2 \Sigma_j^{(3)} \right) \quad j \neq x \quad (12)$$

$$f_{x;x} = \sqrt{1 - \sum_{j \neq x} |f_{x;j}|^2} \quad (13)$$

where S_{xj} uses $\lambda_x - \lambda_j$ (all evaluated at u_0).

Validity check: $\sum_{j \neq x} |f_{0;j}|^2 < 1$. If violated, $|u_0|$ is insufficient.

5 Observables

5.1 Transition probabilities

For a given initial eigenvector x :

$$P_{x \rightarrow i} = |f_i(u_f)|^2$$

5.2 Transition matrix

Running the ODE for each $x \in \{0, \dots, N-1\}$ produces the $N \times N$ matrix:

$$\mathbf{P} = \begin{pmatrix} P_{0 \rightarrow 0} & P_{0 \rightarrow 1} & \cdots & P_{0 \rightarrow N-1} \\ P_{1 \rightarrow 0} & P_{1 \rightarrow 1} & \cdots & P_{1 \rightarrow N-1} \\ \vdots & & \ddots & \vdots \\ P_{N-1 \rightarrow 0} & P_{N-1 \rightarrow 1} & \cdots & P_{N-1 \rightarrow N-1} \end{pmatrix}$$

Unitarity: each row sums to 1. This matrix is the primary output for parameter sweeps.